R109/A_tied_late | Method-conditioned nodes and actual latency

Complete standalone model including candidate node refinement; unequal parameter counts and no equal-FLOPs claim.

Problem	Instances	Milliseconds / instance
TSP	128	1.500991
CVRP	128	2.208654
ATSP	128	0.164397
OVRP	128	0.178952
VRPB	128	0.122584
VRPL	128	0.179708
VRPTW	128	0.179586
OVRPTW	128	0.179244
OVRPB	128	0.122303
OVRPL	128	0.179064
VRPBL	128	0.122846
VRPBTW	128	0.122220
VRPLTW	128	0.179463
OVRPBL	128	0.122593
OVRPBTW	128	0.122363
OVRPLTW	128	0.179478
VRPBLTW	128	0.123016
OVRPBLTW	128	0.122484

Total trainable parameters: 2,305,033; candidate-conditioned nodes, tied two-layer graph weights

Optional R94B: not_requested

R109 refines each candidate node graph through two layers on shared original H4 before pooling. A_tied_late ties these weights; B_private_late has 21 banks keyed by global method ID, never candidate column, winner or outcome. Both retain candidate conditioning, original two joint layers, score MLP/maximin and original R77 CE/comparison/risk/two-forward RDropObjective. A has 2,305,033 trainables; B has 10,220,553. The original 1,842,953-parameter base is fresh matched, and all late branches start from identical original-layer prototypes. The initial eval function is identical, but complete state hashes/dimensions are not identical. This is not parameter/FLOP-matched, a routing experiment, or an isolated capacity-only control. Original R88B physical TSP node0 role, methodID+34 and geometry/distance remain; 57,600 updates, logical batch1024, seed2. No new route/path fields, labels, donor, teacher, context, targets, calibration, quotas or auxiliary loss. R89A AdamW, R89B Muon and R84A are saved historical references, not rerun or newly parameter/FLOP-matched controls; R94B is optional. NSS uses saved historical R79 weights, not retrained on new90k. Fixed36 true rescues/losses, false picks and aligned population net costs/regret are separate from selection frequency. Native/FP64 ind and legal pools; TRAIN-fitted SBS; ALL task-equal. TEST is development-exposed; one whole-model full18 VAL checkpoint per arm for full TRAIN/VAL/TEST/LIB; no TEST/LIB reselection. Actual standalone GPU-resident-input timing includes complete method-conditioned node refinement, excludes dataset I/O, CPU transfer and solvers, and is not end-to-end dataset latency. Microcaps/eval batch come from the main-witness locked runner, not this reporter. No routing summary is required. No online W&B equality/availability is asserted.

R109/A_tied_late | Same selected model: full TRAIN

Diagnostic only; TRAIN was never used to select the checkpoint; full TRAIN-fitted SBS.

Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)
ALL	71.4818	66.0332	11.388574	11.400702	11.345073	-0.2035	+0.3813	0.3893
TSP	46.1877	32.8617	8.003843	8.023959	7.970215	-0.2507	+0.4219	0.4178
CVRP	48.1358	18.1284	18.039791	18.151987	17.946816	-0.6181	+0.5181	0.6789
ATSP	76.7086	46.9210	1.639892	1.678679	1.633938	-2.3106	+0.3644	0.3656
OVRP	41.8074	34.2420	7.443315	7.453531	7.397676	-0.1371	+0.6169	0.6292
VRPB	80.8543	80.2765	9.354826	9.357264	9.319342	-0.0261	+0.3808	0.3873
VRPL	38.3667	31.5309	12.391994	12.400882	12.344503	-0.0717	+0.3847	0.3909
VRPTW	80.4494	80.2407	18.812223	18.813703	18.757954	-0.0079	+0.2893	0.3031
OVRPTW	83.9556	83.8025	10.117900	10.118640	10.093220	-0.0073	+0.2445	0.2388
OVRPB	88.4926	88.3580	6.362634	6.363230	6.346658	-0.0094	+0.2517	0.2587
OVRPL	42.2593	34.4716	7.443946	7.453219	7.398483	-0.1244	+0.6145	0.6245
VRPBL	80.0790	79.4593	9.400342	9.403115	9.363196	-0.0295	+0.3967	0.3980
VRPBTW	79.7025	79.4914	19.730488	19.733549	19.642712	-0.0155	+0.4469	0.4310
VRPLTW	80.5481	80.3827	18.768051	18.769959	18.714366	-0.0102	+0.2869	0.3006
OVRPBL	88.6383	88.5556	6.358650	6.358857	6.342513	-0.0033	+0.2544	0.2622
OVRPBTW	83.3654	83.2432	10.623416	10.624446	10.586225	-0.0097	+0.3513	0.3257
OVRPLTW	83.8790	83.7321	10.110367	10.111265	10.085768	-0.0089	+0.2439	0.2398
VRPBLTW	79.7963	79.5630	19.768165	19.770814	19.680364	-0.0134	+0.4461	0.4287
OVRPBLTW	83.4469	83.3370	10.624494	10.625544	10.587365	-0.0099	+0.3507	0.3267

ALL task-equal; ratios of mean costs differ from per-instance actual regret.

R109/A_tied_late | Fixed target36 vs R89A

True rescues/losses, false picks and aligned net cost; fixed targets, no reselection.

Split	Winners	Rescued	Lost	New false	Net cost delta
TRAIN	83279	538	75	450	-0.000538392
VAL	4625	20	3	27	-0.000198650
TEST	4598	20	5	33	0.000010360

Split	Top1 delta pp	Corrected	Harmed	Cost delta	Regret delta pp
TRAIN	0.237311	23099	19639	-0.000538392	-0.005407
VAL	0.051852	1179	1137	-0.000198650	-0.000968
TEST	0.153086	1227	1103	0.000010360	-0.000199

Historical only, not retrained/matched; R89A AdamW, R89B Muon, R84A AdamW.

Fixed-target-only deltas do not replace the full18 population net cost/regret above.

R109/A_tied_late | Fixed target36 vs R89B

True rescues/losses, false picks and aligned net cost; fixed targets, no reselection.

Split	Winners	Rescued	Lost	New false	Net cost delta
TRAIN	83279	245	475	308	0.000497934
VAL	4625	8	28	21	0.000339330
TEST	4598	10	24	22	0.000325394

Split	Top1 delta pp	Corrected	Harmed	Cost delta	Regret delta pp
TRAIN	-0.209328	20165	23217	0.000497934	0.005660
VAL	-0.145679	1142	1260	0.000339330	0.004037
TEST	-0.034568	1169	1197	0.000325394	0.004391

Historical only, not retrained/matched; R89A AdamW, R89B Muon, R84A AdamW.

Fixed-target-only deltas do not replace the full18 population net cost/regret above.

R109/A_tied_late | Fixed target36 vs R84A

True rescues/losses, false picks and aligned net cost; fixed targets, no reselection.

Split	Winners	Rescued	Lost	New false	Net cost delta
TRAIN	83279	558	61	459	-0.000586049
VAL	4625	23	2	29	0.000022136
TEST	4598	22	4	35	-0.000047198

Split	Top1 delta pp	Corrected	Harmed	Cost delta	Regret delta pp
TRAIN	0.237929	24263	20794	-0.000586049	-0.005904
VAL	0.017284	1241	1227	0.000022136	0.001219
TEST	0.149383	1291	1170	-0.000047198	-0.000810

Historical only, not retrained/matched; R89A AdamW, R89B Muon, R84A AdamW.

Fixed-target-only deltas do not replace the full18 population net cost/regret above.